The first exam of mathematics (evenning studies)

Q1) The third term is $(-18)$and the sixth term is 9216. Find the first nd nth term.

Q2) Find the first four parital sums and the nth parital sum of the sequence an

$a_n = \sqrt{n} - \sqrt{n+1}$

Q3)1) Howmany terms of the arithmetic sequence 5, 7, 9, ... must be added to get 572?

2) Write the sum using sigma notation $1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots + 100x^{99}$, and howmany terms of the arithmetic sequence .

Q4)1) Find the sum $\sum_{k=1}^{16} 64(\frac{3}{2})^{k-1}$.

2) Determine weather the in finite geometric series is convergent or divergent, if itis gonvergent, find the sum

First Exam in mathematics for computing
University of Basrah
College of Computer Science and Information Technology
Department of Computer Science
2022-2023 / First Semester

Date: 21/2/2023

Exam Duration: one hour

Q1) Choose any three of the following questions: (4 marks for each one)

1. A system of equations is given , $\begin{cases} 2x - y = 7 \\ -x + 2y = -5 \end{cases}$

Solving a System Using a Matrix Inverse by

(a) Write the system of equations as a matrix equation.

(b) Solve the system by solving the matrix equation.

2. Find the value of x if $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = 0$

3. If $\begin{bmatrix} a+4 & 3b \\ 8 & -6 \end{bmatrix} = \begin{bmatrix} 2a+2 & b^2+2 \\ 8 & b^2-5b \end{bmatrix}$, then find the value of ***a and b***

4. If $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$, then find $A + A^T + A^{-1}$

Q2) Using Cramer's Rule to solve the following linear system (8 marks)

$2x + 2y + z = 1$

$x - y + 6z = 21$

$3x + 2y - z = -4$

University of Basrah
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B

Date: 28/12/2022

Exam Duration: one hour

Q1) Fill in the blank for the following questions: (2 marks for each one)

1. If $\sum_{k=1}^{5} 3\left(\frac{1}{2}\right)^{k-1}$, then the sum is ....
2. The sum of the first 50 positive integers is ....
3. $\sum_{k=1}^{40} 5 =$..........
4. The missing term of the arithmetic sequence 22, [ ], 34, ... is ....

Q2) Decide if the following infinite geometric series converge or diverge. If it is convergent, find the sum : (2 marks for each one)

1. $\frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \cdots$
2. $\frac{2}{3} - \frac{1}{3} + \frac{1}{6} + \cdots$
3. $\frac{3}{5} + \frac{3}{20} + \frac{3}{80} + \cdots$

Q3) Choose any one of the following questions (6 marks for each one)

A) Express 2.252525... as fraction

B) Find the 100^{th} partial sum of the arithmetic sequence

$2, 5, 8, 11, \ldots$

Good Luke

First Exam in mathematics for computing

University of Basrah

College of Computer Science and Information Technology

Department of Computer Science

2022-2023 / First Semester

Exam Duration: one hour

Date: 23/2/2023

Q1) Choose any three of the following questions: (4 marks for each one)

1. A system of equations is given , $\begin{cases} 5x + 3y = 15 \\ 4x - 2y = 12 \end{cases}$

Solving a System Using a Matrix Inverse by

(a) Write the system of equations as a matrix equation.

(b) Solve the system by solving the matrix equation.

2. If $\begin{vmatrix} 1 & 2 & 1 \\ 2 & x & 2 \\ 3 & 6 & x \end{vmatrix} = 0, \textit{then find the value of } x.$

3. Find the determinant of the matrix $A = \begin{bmatrix} 2 & -1 & 3 & 0 \\ 4 & -2 & 7 & 0 \\ -3 & -4 & 1 & 5 \\ 6 & -6 & 8 & 0 \end{bmatrix}$

4. If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ then show that $A^2 - 4A + 7I = O$, where I is an identity matrix and O is the zero matrix.

Q2) Using Cramer's Rule to solve the following linear system

$2x + 3y - z = 8$

$x + y + z = 2$

$3x - 4y + 5z = -3$

$D = 19$

$D_x = 38$

$D_y = 19$

$D_z = -19$

First Exam in mathematics for computing
University of Basrah
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2022-2023 / First Semester

Exam Duration: one hour

Date: 12/2/2023

Q1) Choose any three of the following questions: (4 marks for each one)

1. A system of equations is given , $\begin{cases} -3x + 10y = 3 \\ x - 3y = -3 \end{cases}$

Solving a System Using a Matrix Inverse by

(a) Write the system of equations as a matrix equation.

(b) Solve the system by solving the matrix equation.

2. If $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$, then find $B^{-1}A^{-1}$

3. If $A = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 \\ -6 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$ $then\ find\ \frac{|B|}{|A|} + \frac{|A|}{|C|}$

4. If $\begin{vmatrix} 8 & -5 & 1 \\ 5 & x & 1 \\ 6 & 3 & 1 \end{vmatrix} = 4, then\ find\ the\ value\ of\ x.$

Q2) Using Cramer's Rule to solve the following linear system

$x - 2y + z = 4$

$x - y - z = --2$

$2x + y + z = 5$

The first exam of mathematics (evenning studies)

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First Exam in mathematics for computing
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2022-2023 / First Semester

Date: 12/2/2023

Exam Duration: one hour

Q1) Choose any three of the following questions: (4 marks for each one)

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Solving a System Using a Matrix Inverse by

(a) Write the system of equations as a matrix equation.

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3. If $A = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 \\ -6 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$ $then\ find\ \frac{|B|}{|A|} + \frac{|A|}{|C|}$

4. If $\begin{vmatrix} 8 & -5 & 1 \\ 5 & x & 1 \\ 6 & 3 & 1 \end{vmatrix} = 4, then\ find\ the\ value\ of\ x.$

Q2) Using Cramer's Rule to solve the following linear system

$x - 2y + z = 4$

$x - y - z = --2$

$2x + y + z = 5$

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Date: 21/2/2023

Exam Duration: one hour

Q1) Choose any three of the following questions: (4 marks for each one)

1. A system of equations is given , $\begin{cases} 2x - y = 7 \\ -x + 2y = -5 \end{cases}$

Solving a System Using a Matrix Inverse by

(a) Write the system of equations as a matrix equation.

(b) Solve the system by solving the matrix equation.

2. Find the value of x if $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = 0$

3. If $\begin{bmatrix} a+4 & 3b \\ 8 & -6 \end{bmatrix} = \begin{bmatrix} 2a+2 & b^2+2 \\ 8 & b^2-5b \end{bmatrix}$, then find the value of $\boldsymbol{a}$ *and* $\boldsymbol{b}$

4. If $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$, then find $A + A^T + A^{-1}$

Q2) Using Cramer's Rule to solve the following linear system (8 marks)

$2x + 2y + z = 1$

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$3x + 2y - z = -4$

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2021-2022 / First Semester

Exam Duration: one hour Date: 17/1/2023

Q1) Choice the correct answer : (2 marks for each one)

1. Write a recursive for the sequence 15,26, 92, 180,... then find the next term

a) $a_n = 2a_n - 4$, where $a_1 = 15, 356$ b) $a_n = 2a_{n-1} - 4$, where $a_1 = 15, 356$

c) $a_n = 4 + 11.2^{n-1} - 4$, where $a_1 = 15, 356$ d) $a_n = 3a_{n-1} - 19$, where $a_1 = 15, 356$

2. What is the formula for the nth term of the sequence 54, 18, 6, ... ?

a) $a_n = 6\left(\frac{1}{3}\right)^n$ b) $6\left(\frac{1}{3}\right)^{n-1}$ c) $a_n = 54\left(\frac{1}{3}\right)^n$ d) $a_n = 54\left(\frac{1}{3}\right)^{n-1}$

3. If $S_n = 86185, r = 2, n = 10$ then a_1

a)8609.50 b) 252.25 c) 84.25 d)7834.00

4. The sum of the first 180 positive integers is

a)16920 b)16290 c)19260 d)19620

Q2) Decide if the following infinite geometric series converge or diverge. If it is convergen find the sum : (2 marks for each one)

1. $1 + 0.1 + 0.01 + 0.001 + 0.0001 + \cdots$
2. $2 + \frac{4}{3} + \frac{8}{9} + \frac{16}{27} \dots$
3. $1 - \frac{3}{4} + \frac{9}{16} \dots$
4. $\sum_{n=1}^{\infty} 5(0.8)^{n-1}$

Q3) Choose any one of the following questions (6 marks for each one)

A) Express 32.32323232... as fraction

B) Theater Seating An architect designs a theater with **15**seats in the first row, **18** in the second, **21** in the third, and so on. If the theater is to have a seating capacity of **870**, many rows must the architect use in his design ?

Good Luke

First Exam in mathematics for computing
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2022-2023 / First Semester

Duration: one hour

Date: 1/2/202

Q1) fill the blank with the correct answer (choose 5 only) (2 marks for each one)

1. The terms of the sequence are $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \ldots$, then $S_5 =$ ………
2. If $a = \frac{5}{2}$ and $r = -\frac{1}{2}$, then the partial sum of the geometric sequence is ……….
3. If $\sum_{k=1}^{6} 3\left(\frac{1}{2}\right)^{k-1}$, then the sum is ………
4. The sum of the infinite geometric series $1 + \frac{2}{5} + \frac{4}{25} + \frac{8}{125} + \frac{16}{625} + \cdots$ is ………
5. The missing term of the geometric sequence 45, [ ], 1620 is ………..
6. 7. The sum of the numbers, 1, 2, 3, 4, 5, 6, 7, …, 2000 is …………..

Q2) Choose the correct answer for the following questions: (2 marks for each one)

1. The missing term of the geometric sequence 45, [ ], 1620, … is

   a) 270 b) 720 c) 270 d) 225

2. The sum of the infinite geometric series, $\frac{1}{5} + \frac{3}{25} + \frac{9}{125} + \frac{27}{625} + \cdots$ is

   a) diverges b) $\frac{5}{7}$ c) $\frac{5}{6}$ d) $\frac{1}{2}$

3. Find the 50th term of the sequence, 5, −2, −9, −16, …

   a) 352 b) −343 c) −338 d) 331

4. The missing term of the arithmetic sequence, 31, [ ], 47

   a) 37 b) 39 c) 41 d) 35

5. Find the sum of the infinite geometric series, $\frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \cdots + \frac{3}{10^n} + \cdots$ is

   a) $\frac{1}{2}$ b) $\frac{1}{4}$ c) $\frac{1}{3}$ d) $\frac{2}{3}$

Good Luke